

2027 ODI WORLD CUP

Team & Venue Intelligence Analysis

Data Analyst Portfolio Project

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Python • Pandas • JSON • Power BI • DAX

1. Executive Summary

This project develops a data-driven team and venue intelligence solution for the 2027 ODI World Cup project scope. The analysis converts ball-by-ball ODI JSON data into team–venue performance measures, applies a weighted scoring framework, and presents the resulting recommendations through an interactive Power BI dashboard.

The final analytical scope contains 14 project teams, 12 project venues, and 168 possible team–venue combinations. At the team recommendation level, 11 of 14 recommendations are supported by historical venue data and 3 use the team fallback approach. The strongest recommendation is Pakistan at Queens Sports Club with a score of 87.10.

2. Project Snapshot

Metric	Value
Project teams	14
Project venues	12
Team–venue combinations	168
Historical recommendation coverage	78.57%
Fallback recommendation coverage	21.43%
Highest recommendation score	87.10
Highest scoring team	Pakistan
Top recommended venue	Queens Sports Club
Qualifying ODI matches in master dataset	153

The Python master dataset contains 153 qualifying match-level records. The dashboard's Historical Matches KPI is based on team–venue summary observations, so a match can contribute to both participating teams.

3. Business Objective

- Identify historically favorable venues for each target team.
- Compare team batting and bowling performance across selected venues.
- Create a consistent relative scoring framework for venue suitability.
- Provide fallback recommendations where historical team–venue observations are unavailable.
- Communicate the analysis through an interactive Power BI dashboard.

4. Data Source and Scope

The Python workflow reads 2,565 men's ODI JSON files and filters them to men's ODI matches from 1 January 2015 onward at the selected project venues.

The analytical scope contains 14 project teams and 12 project venues. Venue-name variants are standardized before aggregation.

Project teams: Australia, Bangladesh, England, India, Ireland, New Zealand, Pakistan, Scotland, South Africa, Sri Lanka, West Indies, Zimbabwe, Namibia, Afghanistan.

Project venues: Wanderers Stadium, SuperSport Park, Newlands, Kingsmead, St George's Park, Mangaung Oval, Boland Park, Buffalo Park, Harare Sports Club, Queens Sports Club, Mosi-oa-Tunya International Cricket Stadium, Namibia Cricket Ground.

5. Data Preparation

- JSON ingestion: Read the 2,565 JSON match files and extract match ID, date, venue, teams, winner, gender, and match type.
- Scope filtering: Keep men's ODI matches dated from 2015-01-01 onward and matches associated with the standardized project venues.
- Venue standardization: Map historical venue-name variations such as different Wanderers, Newlands, SuperSport Park, St George's Park, and other naming forms to a common venue label.
- Batting extraction: Parse each delivery to capture batter runs, balls faced, fours, sixes, and dismissals; aggregate these into player-innings and team–venue measures.
- Bowling extraction: Parse legal deliveries, runs charged to bowlers, and bowler-credited wickets; aggregate these into team–venue bowling measures.
- Match results: Create team-level venue records containing matches, wins, losses, and win percentage.
- Master analytical layer: Merge batting, bowling, and result summaries by team and standardized venue.

6. Scoring Methodology

Batting score

Component	Weight
Runs per innings	40%
Strike rate	30%
Fours	15%
Sixes	15%

Bowling score

Component	Weight
Wickets per match	40%
Economy	25%
Bowling average	20%
Bowling strike rate	15%

The final ranking pipeline combines batting and bowling scores equally: 50% batting and 50% bowling. The notebook also explored a win-percentage-based venue score earlier, but that component is not part of the final venue ranking construction.

7. Fallback Recommendation Logic

All $14 \times 12 = 168$ team–venue combinations are generated. Historical scores are used where available. Missing combinations receive the team's historical weighted venue score; where that is unavailable, a neutral baseline derived from available historical venue scores is used.

The final data source distinguishes Historical Venue Data from Team Fallback.

8. Key Findings

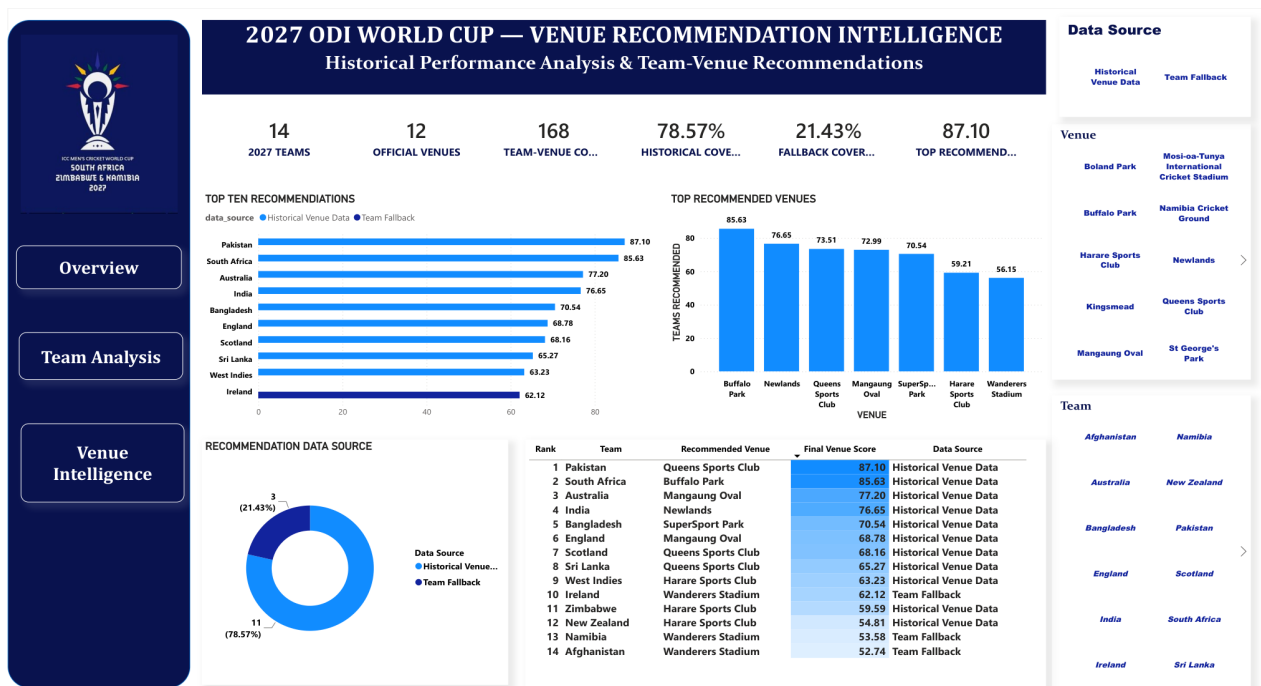
- Pakistan has the highest final recommendation score at 87.10, with Queens Sports Club as the recommended venue.
- South Africa is second in the final recommendation ranking at 85.63, with Buffalo Park recommended.
- Australia is third at 77.20, with Mangaung Oval recommended.
- India is fourth at 76.65, with Newlands recommended.
- The team-level recommendation output contains 11 historical-data recommendations and 3 fallback recommendations, corresponding to 78.57% and 21.43%.

- Across the 12 project venues, Harare Sports Club has the largest historical match volume in the dashboard summary, followed by Queens Sports Club and SuperSport Park.
- The project contains 168 possible team–venue combinations. The dashboard's coverage analysis identifies 68 combinations with historical data and 100 requiring fallback coverage, corresponding to 40.5% and 59.5% respectively.

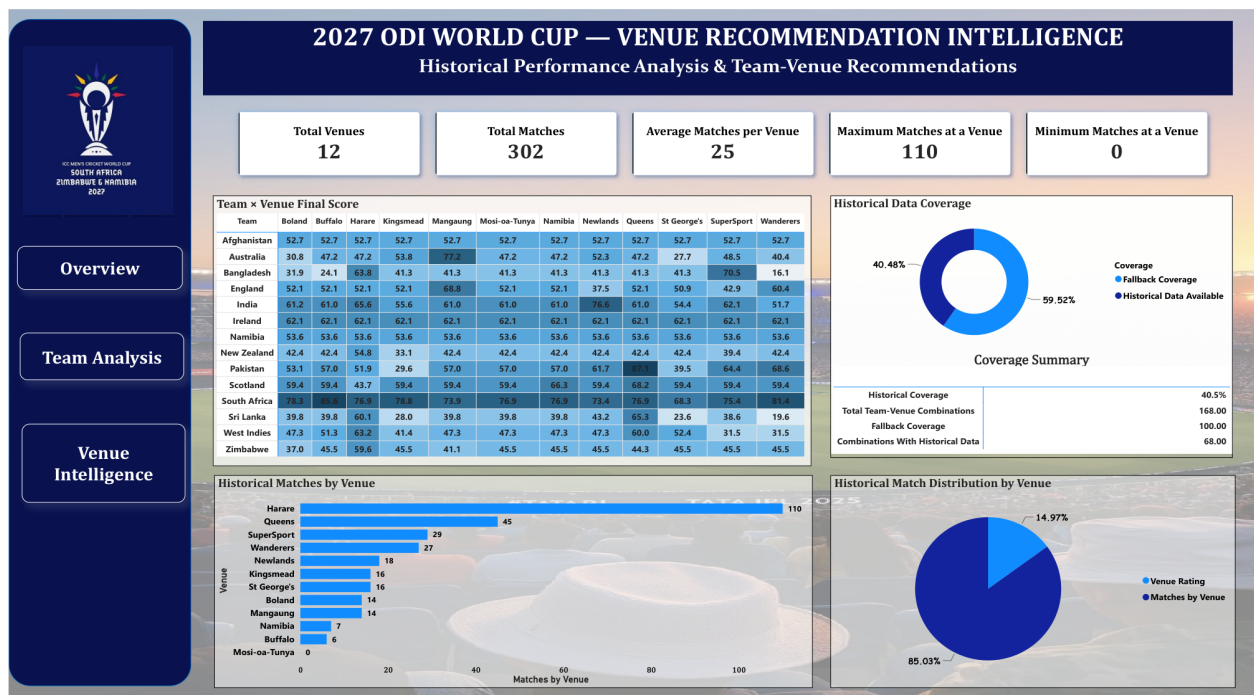
9. Power BI Dashboard

- Page 1 — Overview: Summarizes the recommendation ranking, recommended venues, recommendation data sources, and project-level KPIs.
- Page 2 — Team Analysis: Provides team selection, venue ranking, score breakdown, top recommended venues, recommendation summary, and historical venue KPIs.
- Page 3 — Venue Recommendation Intelligence: Provides the team × venue final-score matrix, historical/fallback coverage, historical match volume by venue, and venue-level intelligence.

Dashboard page 1







10. Technical Architecture

- Source layer: ODI JSON match files.
- Python processing layer: JSON parsing, filtering, venue standardization, batting extraction, bowling extraction, and match-result aggregation.
- Analytical tables: team-venue batting, team-venue bowling, team-venue results, venue scores, final venue ranking, final recommendations, and historical team-venue summary.
- BI layer: Power BI data model with dimension tables such as DimTeam and DimVenue, plus recommendation and historical summary tables.
- Presentation layer: interactive slicers, KPI cards, ranking charts, score breakdowns, matrix/heatmap, coverage visuals, and navigation buttons.

11. Limitations and Interpretation

- The recommendations are analytical suitability scores, not guaranteed match outcomes.
- Historical availability is uneven across teams and venues; fallback scores are used to avoid leaving combinations unrated.
- Rank-based normalization makes the scores relative to the analytical population rather than absolute measures of cricket quality.
- The dashboard's 302 historical-match KPI is a team-venue summary total and can count a match for each participating team; it should not be interpreted as 302 unique matches.
- The project venue list is the defined analytical scope used for this project; the report does not independently verify tournament scheduling or official fixture allocation.

12. Conclusion

The project demonstrates an end-to-end data analyst workflow: raw JSON data is transformed into structured analytical datasets, performance measures are calculated with transparent weighting rules, missing historical coverage is handled explicitly, and the results are communicated through an interactive Power BI dashboard.

The solution demonstrates practical skills in Python data preparation, analytical scoring, data modeling, DAX, Power BI visualization, and business-oriented storytelling.